World Population and CO2 Analysis

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2026-05-12

#Exploratory Data Analysis (EDA) # Load CSV files population_data <- read.csv("Assignment/world_population.csv") co2_data <- read.csv("Assignment/CO2_emission.csv")

View Data sets

head(population_data) head(co2_data)

Load Libraries

library(dplyr)

Display Variable names

colnames(population_data) rownames(population_data)

Top 5 rows

head(population_data, 5)

bottom 10 rows

tail(population_data, 10)

shape of Data set

dim(population_data)

Check duplicate rows

Duplicate_rows <- sum(duplicated(population_data)) head(Duplicate_rows)

Remove Duplicate

population_data <- distinct(population_data) head(population_data)

missing value in each column

missing_value_data <- colSums(is.na(population_data)) head(missing_value_data)

Structure of Dataset

str(population_data)

Box Plot for Quantitative Variables

boxplot(population_data[sapply(population_data, is.numeric)], main = "BoxPlot for Numeric Variables", las = 2) # Replace Missing numeriv values with Median population_data <- population_data %>% mutate(across(where(is.numeric), ~ifelse(is.na(.), median(., na.rm = TRUE), .)))

Create the function

remove_outliers <- function(x) {

Q1 <- quantile(x, 0.25, na.rm = TRUE) Q3 <- quantile(x, 0.75, na.rm = TRUE) IQR <- Q3 - Q1

x[x < (Q1 - 1.5 * IQR) | x > (Q3 + 1.5 * IQR)] <- NA

return(x) }

Apply function to numeric columns

population_data[sapply(population_data, is.numeric)] <- lapply(population_data[sapply(population_data, is.numeric)], remove_outliers)

summary(population_data)

Generating a New Variable Using World Population Dataset

View column names first

colnames(population_data)

Create projected population for 2030

population_data <- population_data %>% mutate(Population_2030 = X2022.Population * exp(Growth.Rate * 8)) # View first rows head(population_data)

Summary of projected population

summary(population_data$Population_2030)

Check actual column names

names(population_data)